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FIPS 203/204/205 Compliant Cryptographic Architecture for Naval Information Dominance

TECHNICAL WHITE PAPER

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Abstract

We present a comprehensive cryptographic architecture implementing all three NIST post-quantum standards—ML-DSA (FIPS 204), ML-KEM (FIPS 203), and SLH-DSA (FIPS 205)—integrated into a production blockchain consensus system with hybrid classical/quantum security. The Quasar consensus protocol achieves sub-second finality through dual-certificate blocks combining BLS12-381 aggregate signatures with Ringtail lattice-based threshold signatures, providing an attack window of $\leq$50 ms. We detail six threshold multi-party computation schemes (CGGMP21, FROST, LSS, BLS, SR25519, TFHE) operating over a zero-copy wire protocol, GPU-accelerated cryptographic operations on Metal and CUDA, fully homomorphic encryption via a dedicated virtual machine, and a comprehensive key management system with Shamir secret sharing, HSM integration, and HIPAA/SOX/SEC compliance controls. The architecture provides defense-in-depth against both current and quantum adversaries, with 17 formal cryptographic proofs supporting security claims.

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1 Introduction

The transition to post-quantum cryptography represents the most significant cryptographic migration in the history of information security. The National Security Agency's CNSA 2.0 guidance mandates migration to quantum-resistant algorithms by 2030 for national security systems, while adversaries are presumed to be conducting harvest-now-decrypt-later (HNDL) attacks today.

This paper presents a cryptographic architecture that has moved beyond theoretical compliance to production deployment. Every component described herein is implemented, tested, and operational—not a research proposal but an engineering reality.

1.1 Scope of NIST Compliance

Standard	Algorithm	Purpose	Integration Points
FIPS 203	ML-KEM-768	Key encapsulation	RNS, ZAP, HPKE, KMS
FIPS 204	ML-DSA-65	Digital signatures	Consensus, RNS, ZAP, EVM
FIPS 205	SLH-DSA	Hash-based signatures	Long-term anchors, audit

Table 1: NIST PQ standard implementation coverage.

2 Post-Quantum Algorithm Implementations

2.1 ML-DSA: Module-Lattice Digital Signatures (FIPS 204)

ML-DSA provides EU-CMA (Existential Unforgeability under Chosen Message Attack) security under the hardness of Module-LWE and Module-SIS problems.

Mode	Public Key	Secret Key	Signature	Security
ML-DSA-44	1,312 B	2,560 B	2,420 B	NIST Level 2
ML-DSA-65	1,952 B	4,032 B	3,309 B	NIST Level 3
ML-DSA-87	2,592 B	4,896 B	4,627 B	NIST Level 5

Table 2: ML-DSA parameter sets.

Definition 2.1 (EU-CMA Security of ML-DSA)**.** For any probabilistic polynomial-time adversary $\mathcal{A}$ making at most q_s signing queries, the probability of producing a valid forgery is:

$$\Pr[\text{EU-CMA}_{\mathcal{A}}^{\text{ML-DSA}} = 1] \leq \text{negl}(\lambda)$$

under the Module-LWE assumption with parameters $(k, \ell, n = 256, q = 8380417)$.

Implementation: Cloudflare CIRCL library (v1.6.1) with CGO optimization and pure-Go fallback. Available as EVM precompile at address `0x0601` with gas cost 75,000–150,000 depending on mode.

2.2 ML-KEM: Module-Lattice Key Encapsulation (FIPS 203)

ML-KEM provides IND-CCA2 security via the Fujisaki-Okamoto transform applied to Module-LWE encryption.

Mode	PK	SK	CT	SS	Security
ML-KEM-512	800 B	1,632 B	768 B	32 B	Level 1
ML-KEM-768	1,184 B	2,400 B	1,088 B	32 B	Level 3
ML-KEM-1024	1,568 B	3,168 B	1,568 B	32 B	Level 5

Table 3: ML-KEM parameter sets (PK: public key, SK: secret key, CT: ciphertext, SS: shared secret).

2.3 SLH-DSA: Stateless Hash-Based Signatures (FIPS 205)

SLH-DSA provides quantum-safe signatures with no algebraic structure assumptions—security relies solely on the second-preimage resistance of the underlying hash function.

- 12 parameter sets: SHA2/SHAKE × 128/192/256-bit × small/fast variants.
- Signature sizes: 7,856–49,856 bytes (tradeoff: signature size vs. verification speed).
- EVM precompile at `0x0602`, gas 50,000–250,000.
- Primary use: long-term audit anchors where algebraic assumptions may not hold for decades.

3 Hybrid Post-Quantum Consensus: Quasar

3.1 Dual-Certificate Architecture

Quasar is a hybrid consensus protocol that produces blocks certified by both classical (BLS12-381) and post-quantum (Ringtail ML-DSA-65) signatures:

Algorithm 1 Quasar Dual-Certificate Block Finalization

Phase I — Classical Finality (target: 500 ms)
Validators sample k peers (mainnet $k = 21$)
Accumulate confidence: β_1 for preference, β_2 for decision
Aggregate BLS12-381 signatures into 96-byte certificate
Block finalized with classical certificate

Phase II — Quantum Finality (target: 3 s bundle)
Bundle 6 BLS-certified blocks into Merkle tree
Each validator computes Ringtail share (Round 1)
Exchange shares, finalize threshold signature (Round 2)
Quantum bundle certified with ML-DSA-65 lattice certificate (∼3 KB)

Block Header: $\mathrm{BLSAgg}_{96\mathrm{B}} \| \mathrm{PQCert}_{3\mathrm{KB}} \| \text{HMAC-SHA256(both)}$

3.2 Attack Window Analysis

Theorem 3.1 (Quasar Quantum Safety)**.** *An adversary with a quantum computer must forge a BLS12-381 aggregate signature within the 50 ms window between classical finality and quantum bundle creation. Even with Shor's algorithm, real-time BLS forgery on current quantum hardware requires wall-clock time exceeding $O(2^{128})$ quantum gates on current architectures, making real-time exploitation infeasible.*

3.3 Epoch-Based Key Rotation

Ringtail threshold keys rotate on epoch boundaries:

- **MinEpochDuration**: 10 minutes (rate-limited to prevent churn).
- **MaxEpochDuration**: 1 hour (maximum key age).
- **HistoryLimit**: 6 epochs (≈1 hour) for cross-epoch verification.
- **QuantumCheckpointInterval**: 3 seconds (frequent PQ anchors).

3.4 Consensus Protocol Family

Quasar orchestrates 10 sub-protocols:

Protocol	Files	Purpose
Quasar	30	PQ consensus orchestrator
Wave	10	Threshold voting, FPC dynamic selection
Photon	5	K-of-N peer selection with luminance tracking
Focus	5	Confidence tracking, windowed confidence
Prism	9	DAG vertex ordering and cutting
Nova	4	Linear blockchain consensus
Nebula	4	DAG consensus mode
Ray	6	Chain consensus protocol
Horizon	5	Event horizon finality detection
Flare	5	Certificate generation, DAG frontier

Table 4: Quasar consensus protocol family.

4 Threshold Multi-Party Computation

The Lux MPC system provides distributed key generation and signing without ever reconstructing full private keys.

4.1 Supported Schemes

Scheme	Curve/Basis	Use Case	Chains
CGGMP21	secp256k1	Threshold ECDSA	20+
FROST	Ed25519, Ristretto255	EdDSA, SR25519	Polkadot, Solana
LSS	secp256k1	Dynamic resharing	All ECDSA
BLS	BLS12-381	Consensus signing	Lux
SR25519	Ristretto255	Substrate compat	Polkadot
TFHE	Lattice	Threshold FHE	T-Chain

Table 5: MPC threshold signature schemes.

4.2 Distributed Key Generation

Algorithm 2 Verifiable Secret Sharing DKG

Phase 1: Commitment
for each party P_i, $i = 1, \ldots, n$ **do**
  Sample random polynomial $f_i(x)$ of degree $t - 1$
  Broadcast commitment $C_i = g^{f_i(0)}$
  Send share $s_{ij} = f_i(j)$ to party P_j (encrypted P2P)
end for

Phase 2: Verification
for each party P_j **do**
  Verify received shares against commitments
  if share invalid **then**
    File complaint against sender
  end if
end for

Phase 3: Finalization
Public key: $\mathrm{PK} = \prod_{i=1}^{n} C_i$
Each P_j holds share: $\mathrm{sk}_j = \sum_{i=1}^{n} s_{ij}$
Shares encrypted with AES-256 and stored locally

Performance: DKG completes in ∼30 seconds for 3 nodes. Signing requires <1 second for threshold signatures.

Byzantine resilience: $t - 1$ compromised parties reveal no information about the secret key, where $t \geq \lfloor n/2 \rfloor + 1$.

4.3 Consensus-Embedded Transport

MPC messages are transported directly over the ZAP consensus wire protocol:

Listing 1: MPC message types in consensus transport.

```
MsgMPCBroadcast  = 60  // Broadcast to all parties
MsgMPCDirect     = 61  // Point-to-point encrypted
MsgMPCMembership = 62  // Ed25519 PoA validators
MsgMPCResult     = 67  // Signing result
```

This eliminates external dependencies (NATS, Consul, Redis) and provides the same availability guarantees as the consensus layer itself.

5 Fully Homomorphic Encryption

5.1 ThresholdVM Architecture

ThresholdVM provides FHE operations as a native virtual machine on the Lux T-Chain, enabling computation on encrypted data without decryption:

- **Schemes**: BFV (integer arithmetic), BGV (modular arithmetic), CKKS (approximate real), TFHE (Boolean circuits).
- **Threshold decryption**: t-of-n parties must cooperate; no single entity sees plaintext.

- **GPU acceleration**: NTT/iNTT, polynomial multiplication, and TFHE bootstrap operations accelerated via LuxCPP Metal/CUDA kernels.
- **EVM integration**: Precompiles at `0x0700`–`0x0703` (Fheos, ACL, InputVerifier, Gateway).

Definition 5.1 (Semantic Security under LWE)**.** For any PPT adversary $\mathcal{A}$, the advantage in distinguishing encryptions of two chosen messages is:

$$\mathrm{Adv}_{\mathcal{A}}^{\text{IND-CPA}}(\lambda) \leq \mathrm{negl}(\lambda)$$

under the Learning with Errors assumption with parameters (n, q, χ).

5.2 TFHE-KMS Bridge

The TFHE-KMS bridge enables encrypted policy evaluation:

1. Client encrypts sensitive query under FHE public key.
2. ThresholdVM evaluates policy predicate homomorphically.
3. Result (encrypted Boolean) is threshold-decrypted by MPC cluster.
4. Client receives only the policy decision, not the underlying data.

6 GPU-Accelerated Cryptography

LuxCPP provides GPU kernels for cryptographic operations across Metal (Apple Silicon), CUDA (NVIDIA), and WebGPU:

Category	**Operations**
Elliptic Curves	BLS12-381, BN254, secp256k1, Ed25519: point add/double/scalar-mul, pairing, MSM
Post-Quantum	ML-DSA sign/verify, ML-KEM encaps/decaps, NTT/iNTT
ZK Proofs	KZG commitments, polynomial multiplication, Poseidon2 hashing
Hash Functions	SHA3-256/512, BLAKE3, Goldilocks field arithmetic
FHE	TFHE bootstrap, BFV/CKKS modular polynomial ops

Table 6: GPU-accelerated cryptographic operations.

The stable C ABI (`lux-accel/c_api.h`) enables FFI bindings from Go, Rust, and Python.

7 Zero-Knowledge Proof Systems

The ZK precompile suite (`0x0900`–`0x0932`) provides:

- **KZG4844**: Polynomial commitments for data availability (EIP-4844 compatible).
- **Groth16**: Succinct proofs with constant-size verification (∼200K gas).

- **PLONK/fflonk**: Universal setup proofs (∼250K gas).
- **Halo2**: Recursive proofs without trusted setup (∼300K gas).
- **Privacy pools**: Confidential UTXO model with nullifiers and range proofs.
- **ZK rollup**: Batch verification for off-chain computation (∼500K gas).

8 Key Management Architecture

8.1 Hanzo KMS

Enterprise key management with 8 core subsystems:

1. **Shamir Secret Sharing**: GF(2^8) arithmetic for root key distribution (t-of-n threshold).
2. **Transit Engine**: Encryption-as-a-service (AES-256-GCM, ChaCha20-Poly1305, Ed25519, ECDSA, RSA) with key versioning and auto-rotation.
3. **HPKE Key Wrapping**: RFC 9180 hybrid public key encryption for CEK distribution, including ML-KEM-768 + X25519 hybrid mode.
4. **Seal/Unseal Barrier**: AES-256-GCM encryption barrier with auto-unseal via AWS/GCP KMS.
5. **ACL Policy Engine**: Path-based capabilities (deny, CRUD, list, sudo) with template variables and LRU cache.
6. **Token System**: Service, batch, and recovery tokens with HMAC-SHA256 integrity and parent-child revocation cascade.
7. **Dynamic Secrets**: Auto-created temporary credentials for PostgreSQL, MySQL, Redis, MongoDB.
8. **TFHE-KMS Bridge**: FHE integration for encrypted policy evaluation.

8.2 Compliance Controls

- **HIPAA**: Break-glass emergency decryption tokens (time-limited), WORM hash-chained audit logs.
- **SEC 17a-4 / SOX**: 6–7 year retention enforcement, regulator escrow shards, immutable audit.
- **GDPR**: Per-org encryption keys, graceful secret revocation.
- **Cloud logging**: AWS CloudWatch + S3 Object Lock, GCP Cloud Logging + GCS WORM, Azure Monitor.

9 Ring Signatures and Anonymity

9.1 LatticeLSAG: Post-Quantum Ring Signatures

LatticeLSAG extends Linkable Spontaneous Anonymous Group signatures to ML-DSA-65:

- **Anonymity**: Verifier cannot determine which ring member signed.

- **Linkability**: Key images enable detection of double-signing (same signer produces same image).
- **Spontaneous**: Any set of public keys can form a ring without coordination.
- **Post-quantum**: Based on Module-LWE hardness, resistant to Shor's algorithm.
- **Application**: Anonymous reporting, covert intelligence sharing, whistleblower protection.

10 Software-Defined Networking Architecture

The combination of zero-trust overlay networking and hot-reloadable gateway configuration constitutes a software-defined networking (SDN) architecture with clear control plane / data plane separation.

10.1 Control Plane

- **Policy engine**: The Hanzo Gateway applies declarative routing, authentication, and rate-limiting policies defined in KMS-backed configuration. Policy updates propagate to all gateway instances within seconds without connection interruption.
- **Zero-trust controller**: The Hanzo ZT controller maintains the identity fabric—enrolling endpoints, issuing short-lived certificates, and distributing routing tables to edge routers. All control traffic is encrypted with ML-KEM-768 + X25519 hybrid key exchange.
- **Service mesh**: Mutual TLS between all services is enforced by the overlay, with certificate rotation on 15-minute intervals. Services are addressed by identity, not IP, eliminating lateral movement via network scanning.

10.2 Data Plane

- **Overlay transport**: Traffic between endpoints traverses encrypted tunnels (WireGuard or zero-trust fabric), with no reliance on network-layer security. The overlay is transparent to application protocols.
- **Policy-based routing**: Ingress and egress rules are evaluated per-packet based on caller identity, destination service, and request metadata. Routing decisions are made at the edge, not at a central choke point.
- **Hot reconfiguration**: Gateway routing tables, TLS certificates, and access control lists reload without process restart or connection drop, enabling continuous deployment of network policy changes under operational tempo.

This SDN architecture enables network segmentation, microsegmentation, and policy enforcement that adapts in real time to threat conditions—critical for contested electromagnetic environments where network topology may change rapidly.

11 Formal Verification

The architecture is supported by 17 formal cryptographic proofs:

Proof	Claim
proof-crypto-bls	BLS12-381 aggregate signature security
proof-crypto-cggmp21	CGGMP21 threshold ECDSA correctness
proof-crypto-mldsa	ML-DSA EU-CMA reduction to Module-LWE
proof-crypto-tfhe	TFHE semantic security under LWE
proof-crypto-verkle	Verkle tree binding and hiding
proof-protocol-wave	Wave consensus liveness and safety
proof-protocol-nova	Nova linear consensus termination
proof-bridge-teleport	Cross-chain message integrity
proof-warp-delivery	Warp messaging delivery guarantee

Table 7: Selected formal proofs (9 of 17 shown).

12 E2E Post-Quantum Integration Map

Layer	Classical	PQ Hybrid	Status
Consensus finality	BLS12-381	Ringtail ML-DSA-65	Live
RNS mesh transport	X25519	ML-KEM-768 + ML-DSA-65	Live
TLS 1.3 key exchange	X25519	X25519+ML-KEM-768	Live
SessionVM messaging	—	ML-KEM-768 + ML-DSA-65	Live
ZAP protocol	X25519	ML-KEM-768 + ML-DSA-65	Live
KMS encryption	HPKE-X25519	Hybrid HPKE	Implemented
EVM precompiles	ECDSA	ML-DSA, ML-KEM, SLH-DSA	Implemented

Table 8: End-to-end post-quantum integration status.

13 Naval Cyber Operations Scenarios

13.1 Defensive Cyber Operations (DCO)

- **HNDL protection**: All network traffic encrypted with hybrid PQ, rendering recorded ciphertext useless to quantum adversaries.
- **Key compromise recovery**: Epoch-based rotation limits exposure window to 10 minutes; threshold MPC ensures no single key compromise is catastrophic.
- **Anomaly detection**: O11y observability platform provides real-time trace analysis with LLM-powered threat classification.

13.2 Offensive Cyber Operations (OCO)

- **Penetration testing**: 83 specialized AI agents with multi-agent orchestration for automated vulnerability assessment.
- **Red team support**: Ring signatures enable anonymous operation; ZK proofs provide verifiable claims without revealing methods.
- **Mission-risk assessment**: Behavioral analytics platform with cohort analysis and predictive modeling.

13.2.1 Automated Penetration Testing Workflow

The agent-driven penetration testing pipeline executes five sequential phases, each coordinated by purpose-built AI agents operating under MPC-protected identity:

Algorithm 3 Agent-Driven Penetration Testing Pipeline

Phase 1 — Reconnaissance
DNS enumeration, port scanning, service fingerprinting
OSINT collection via 14 specialized reconnaissance agents
Attack surface mapping: subdomains, API endpoints, exposed services

Phase 2 — Vulnerability Scanning
Protocol-specific scanners (HTTP, SSH, TLS, SNMP, SMB)
CVE correlation against NVD and internal vulnerability databases
Configuration audit: default credentials, misconfigurations, weak ciphers

Phase 3 — Exploitation
Exploit selection ranked by CVSS score and operational risk
Sandboxed payload execution with rollback capability
Credential harvesting and privilege escalation attempts

Phase 4 — Lateral Movement
Network pivot discovery via ARP, LLMNR, and service enumeration
Pass-the-hash and token impersonation where applicable
Internal reconnaissance of newly accessible network segments

Phase 5 — Reporting
Automated MITRE ATT&CK technique mapping for every action taken
Evidence chain with cryptographic timestamps (SLH-DSA anchored)
Risk-scored findings with remediation recommendations

13.2.2 Red/Blue Team Automation

Adversarial agent pairs operate in continuous red/blue cycles:

- **Red agents**: Execute the penetration pipeline above, adapting tactics based on observed defenses. Agent orchestration supports 200–300 sequential tool calls per engagement, enabling deep vulnerability chain analysis across multi-hop attack paths.
- **Blue agents**: Monitor defensive telemetry in real time via the O11y observability platform, deploying countermeasures (firewall rules, credential rotation, service isolation) within seconds of detected intrusion.
- **Continuous cycle**: Red findings feed directly into blue defensive posture updates; blue countermeasures are re-tested by red agents in the next iteration, producing a closed-loop hardening process.
- **Scoring**: Each cycle produces an adversarial resilience score—the ratio of blocked vs. successful attack paths—tracked over time to measure defensive improvement.

13.2.3 Vulnerability Chain Analysis

Complex environments require analysis of multi-step attack chains rather than isolated vulnerabilities. Agent-driven chain analysis proceeds as follows:

1. **Graph construction**: Build a directed attack graph from reconnaissance and scanning data, where nodes represent hosts/services and edges represent exploitable transitions.
2. **Path enumeration**: Agents traverse the graph using depth-first search with pruning, executing 200–300 sequential tool calls to probe each link in a candidate chain.
3. **Chain validation**: Each candidate chain is validated end-to-end in a sandboxed environment to confirm exploitability under realistic conditions.
4. **Impact scoring**: Validated chains are scored by the product of individual link probabilities and the value of the terminal asset (crown jewel analysis).

13.2.4 Operator Identity Protection

Offensive operations require strong operator anonymity. LatticeLSAG ring signatures (Section 9) protect operator identity during all phases:

- Every tool invocation, exploit attempt, and reporting action is signed with a ring signature over the operator's unit roster.
- Verifiers confirm that an authorized operator performed the action without learning which specific individual.
- Key images prevent a single operator from filing duplicate or contradictory reports (linkability without identifiability).
- All signatures are post-quantum (ML-DSA-65 based), ensuring attribution protection against future quantum adversaries.

13.2.5 MITRE ATT&CK Integration

Every discovered vulnerability and exploitation step is automatically mapped to the MITRE ATT&CK framework:

- **Tactic coverage**: Agents tag each action with ATT&CK tactics (Initial Access, Execution, Persistence, Privilege Escalation, Defense Evasion, Credential Access, Discovery, Lateral Movement, Collection, Exfiltration, Impact).
- **Technique resolution**: Individual techniques (e.g., T1059 Command and Scripting Interpreter, T1078 Valid Accounts) are recorded with sub-technique granularity.
- **Coverage heat map**: Aggregated results produce a heat map of tested vs. untested ATT&CK techniques, identifying gaps in defensive coverage.
- **Reporting**: Final reports include ATT&CK Navigator layers exportable to standard JSON format for integration with existing SOC tooling.

14 QuantumVM: Quantum Computing Operations

QuantumVM is a dedicated virtual machine on the Lux Q-Chain that provides quantum key operations, quantum algorithm simulation, and hybrid quantum-classical workflow support as native blockchain primitives.

14.1 Post-Quantum Key Operations

QuantumVM exposes key lifecycle management for all three NIST PQ standards:

- **Key generation**: On-chain generation of ML-DSA-65, ML-KEM-768, and SLH-DSA key pairs with entropy sourced from the consensus randomness beacon. Keys are stored in threshold-encrypted form—no single validator holds a complete private key.
- **Verification**: Native transaction verification using PQ signatures, with gas costs calibrated to actual computational overhead (ML-DSA-65 verify: 75,000 gas; SLH-DSA verify: 50,000–250,000 gas depending on parameter set).
- **Key rotation**: Epoch-triggered rotation with configurable policy (time-based, usage-count, or external trigger). Rotation is atomic—old keys remain valid for a configurable grace period (default: 2 epochs) to allow in-flight operations to complete.
- **Certificate issuance**: QuantumVM issues short-lived PQ certificates binding identities to public keys, with certificate transparency logs anchored to the Q-Chain state root.

14.2 Quantum Algorithm Simulation

QuantumVM includes a classical simulation backend for quantum circuits, enabling pre-deployment validation of quantum algorithms:

- **Gate set**: Hadamard, CNOT, Toffoli, Phase, T, Rx/Ry/Rz, SWAP, and custom unitary gates up to 3 qubits.
- **Qubit capacity**: State vector simulation up to 30 qubits (single node) or 40 qubits (distributed across MPC cluster).
- **Noise models**: Depolarizing, amplitude damping, and phase flip channels for realistic fidelity estimation.
- **Deterministic execution**: Given identical inputs, simulation produces identical outputs across all validators—a requirement for consensus over quantum computation results.

14.3 Hybrid Quantum-Classical Workflows

Real-world quantum advantage requires tight integration between classical pre/post-processing and quantum subroutines:

Algorithm 4 Hybrid Quantum-Classical Execution on QuantumVM

1: **Classical preprocessing**: Prepare problem instance (e.g., Hamiltonian coefficients, optimization constraints)
2: **Circuit compilation**: Translate high-level algorithm to native gate set with depth optimization
3: **Quantum execution**: Run circuit on simulation backend (or relay to external QPU via oracle bridge)
4: **Measurement**: Collapse state vector; record measurement outcomes on-chain
5: **Classical postprocessing**: Aggregate results, compute expectation values, update variational parameters
6: **Iteration**: Repeat steps 2–5 for variational algorithms (VQE, QAOA) until convergence

The oracle bridge interface allows QuantumVM to submit circuits to external quantum processing units (QPUs) when available, with results verified by threshold consensus before acceptance.

14.4 EVM Precompile Integration

QuantumVM operations are accessible from EVM smart contracts via precompiles at addresses `0x0800`–`0x080F`:

Address	Operation	Gas
`0x0800`	PQ key generation (ML-DSA-65)	200,000
`0x0801`	PQ key generation (ML-KEM-768)	150,000
`0x0802`	PQ key generation (SLH-DSA)	300,000
`0x0803`	PQ signature verification	75,000–250,000
`0x0804`	Key rotation (atomic swap)	100,000
`0x0808`	Quantum circuit submission	500,000+
`0x0809`	Measurement result retrieval	50,000
`0x080A`	Hybrid workflow orchestration	1,000,000+

Table 9: QuantumVM EVM precompile addresses and gas costs.

15 Conclusion

We have presented a production-grade cryptographic architecture that fully implements all three NIST post-quantum standards with hybrid classical/quantum defense-in-depth. The system provides sub-second consensus finality with dual BLS/Ringtail certificates, six threshold MPC schemes, GPU-accelerated cryptographic operations, fully homomorphic encryption, and comprehensive key management with regulatory compliance.

The architecture is not theoretical—it is deployed, tested, and operational, with 17 formal proofs supporting its security claims. It represents the most comprehensive post-quantum cryptographic infrastructure available for naval information dominance operations.

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